



National Institute of
Diabetes and Digestive
and Kidney Diseases

Peripheral Neuropathy

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What is peripheral neuropathy?

Peripheral neuropathy is a type of nerve damage that typically affects the feet and legs and sometimes affects the hands and arms.

This type of neuropathy is very common. Up to one-half of people with diabetes have peripheral neuropathy.^{1,2}

What causes peripheral neuropathy?

Over time, high blood glucose, also called blood sugar, and high levels of fats, such as [triglycerides](#), in the blood from diabetes can damage your nerves and the small blood vessels that nourish your nerves, leading to peripheral neuropathy.

What are the symptoms of peripheral neuropathy?

If you have peripheral neuropathy, your feet, legs, hands, or arms may feel

- burning
- tingling, like “pins and needles”
- numb
- painful
- weak

You may feel extreme pain in your feet, legs, hands, and arms, even when they are touched lightly. You may also have problems sensing pain or temperature in these parts of your body.



If you have peripheral neuropathy, you may feel burning or tingling, like “pins and needles,” in your feet.

Symptoms are often worse at night. Most of the time, you will have symptoms on both sides of your body. However, you may have symptoms only on one side.

If you have peripheral neuropathy, you might experience:

- changes in the way you walk

- loss of balance, which could make you fall more often
- loss of muscle tone in your hands and feet
- pain when you walk
- problems sensing movement or position
- swollen feet

What problems does peripheral neuropathy cause?

Peripheral neuropathy can cause foot problems that lead to blisters and sores. If peripheral neuropathy causes you to lose feeling in your feet, you may not notice pressure or injuries that lead to blisters and sores. Diabetes can make these wounds difficult to heal and increase the chance of infections. These sores and infections can lead to the loss of a toe, foot, or part of your leg. [Finding and treating foot problems](#) early can lower the chances that you will develop serious infections.

This type of diabetes-related nerve damage can also cause changes to the shape of your feet and toes. A rare condition that can occur in some people with diabetes is Charcot's foot, a problem in which the bones and tissue in your foot are damaged.

Peripheral neuropathy can make you more likely to lose your balance and fall, which can increase your chance of fractures and other injuries. The chronic pain of peripheral neuropathy can also lead to grief, anxiety, and [depression](#) [NIH](#) [↗](#).

How do doctors diagnose peripheral neuropathy?

Doctors diagnose peripheral neuropathy based on your symptoms, family and medical history, a physical exam, and tests. A physical exam will include a neurological exam and a foot exam.

Examination for neuropathy

If you have diabetes, you should get a thorough exam to test how you feel in your feet and legs at least once a year. During this exam, your doctor will look at your feet for signs of problems and check the blood flow and feeling, or sensation, in your feet by

- placing a tuning fork against your great toes and higher on your feet to check whether you can feel vibration
- touching each foot and some toes with a nylon strand to see if you can feel it—a procedure called a monofilament test

- reviewing your gait, or the patterns you make when you walk
- testing your balance

Your doctor may also check if you can feel temperature changes in your feet.

What tests do doctors use to diagnose peripheral neuropathy?

Your doctor may perform tests to rule out other causes of nerve damage, such as a blood test to check for [thyroid](#) problems, [kidney disease](#), or low [vitamin B12](#) [NIH](#) [↗](#) levels. If low B12 levels are found, your doctor will do additional tests to determine the cause. [Metformin](#) [NIH](#) [↗](#) use is among several causes of low vitamin B12 levels. If B12 deficiency is due to metformin, metformin can be continued with B12 supplementation.

How can I prevent the problems caused by peripheral neuropathy?

You can prevent the problems caused by peripheral neuropathy by [managing your diabetes](#), which means managing your blood glucose, blood pressure, and cholesterol. Staying close to your goal numbers can keep nerve damage from getting worse.

If you have diabetes, check your feet for problems every day and [take good care of your feet](#). If you notice any foot problems, call or see your doctor right away.

Remove your socks and shoes in the exam room to remind your doctor to check your feet at every office visit. See your doctor for a foot exam at least once a year—more often if you have foot problems. Your doctor may send you to a [podiatrist](#).

How do doctors treat peripheral neuropathy?

Doctors may prescribe medicine and other treatments for pain.

Medicines for nerve pain

Your doctor may prescribe medicines to help with pain, such as certain types of

- [antidepressants](#) [NIH](#) [↗](#), including
 - tricyclic antidepressants, such as [nortriptyline](#) [NIH](#) [↗](#), [desipramine](#) [NIH](#) [↗](#), [imipramine](#) [NIH](#) [↗](#), and [amitriptyline](#) [NIH](#) [↗](#)

- other types of antidepressants, such as [duloxetine](#) ^{NIH}, [venlafaxine](#) ^{NIH}, [paroxetine](#) ^{NIH}, and [citalopram](#) ^{NIH}
- anticonvulsants—medicines designed to treat seizures—such as [gabapentin](#) ^{NIH} and [pregabalin](#) ^{NIH}
- skin creams, patches, or sprays, such as [lidocaine](#) ^{NIH}

Although these medicines can help with the pain, they do not change the nerve damage. Therefore, if there is no improvement with a medicine to treat pain, there is no benefit to continuing to take it and another medication may be tried.

All medicines have side effects. Ask your doctor about the side effects of any medicines you take. Doctors don't recommend some medicines for older adults or for people with other health problems, such as heart disease.

Some doctors recommend avoiding [over-the-counter pain medicines](#) ^{NIH}, such as acetaminophen and ibuprofen. These medicines may not work well for treating most nerve pain and can have side effects.

Other treatments for nerve pain

Your doctor may recommend other treatments for pain, including

- physical therapy to improve your strength and balance
- a bed cradle, a device that keeps sheets and blankets off your legs and feet while you sleep

Diabetes experts have not made special recommendations about [supplements](#) ^{NIH} for people with diabetes. For safety reasons, talk with your doctor before using supplements or any [complementary or alternative medicines](#) ^{NIH} or medical practices.

References

- [1] Pop-Busui R, Boulton AJ, Feldman EL, et al. Diabetic neuropathy: a position statement by the American Diabetes Association. *Diabetes Care*. 2017;40(1):136–154.
- [2] Izenberg A, Perkins BA, Bril V. Diabetic neuropathies. *Seminars in Neurology*. 2015;35(4):424–430.

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